

Unit 11, Lesson 1: Evaluating Expressions

Benchmark

MA.6.AR.1.3 Evaluate algebraic expressions using substitution and order of operations.

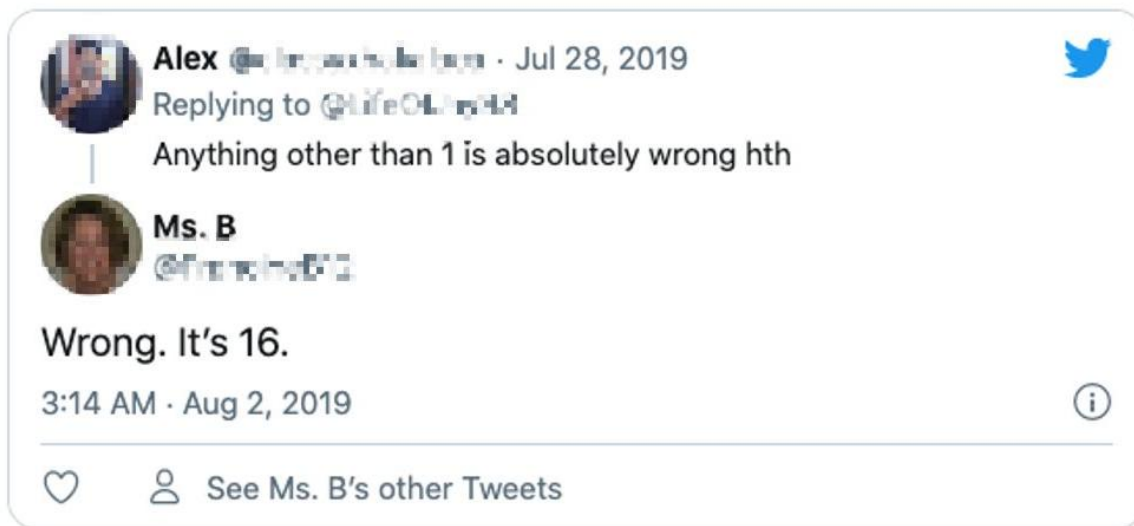
Learning Target

- ☐ I can apply order of operations to an algebraic expression.

11.1.1 Warm-Up: Evaluating Numeric Expressions

This math expression is dividing the internet.

What is the value of $8 \div 2(2 + 2)$?

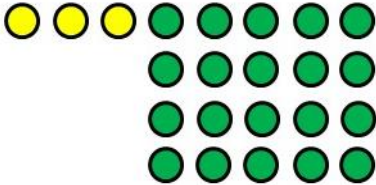
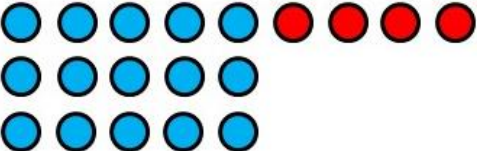
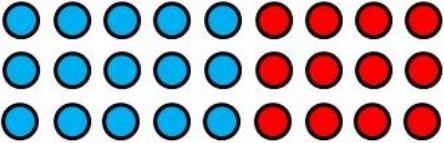
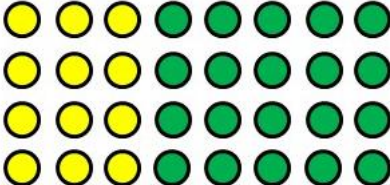


1. Do you agree with Alex or Ms. B? Explain or show why using order of operations.
2. Find the value of the expression. Show your work.

$$4 - 2 \times 6 \div 3 + 7$$

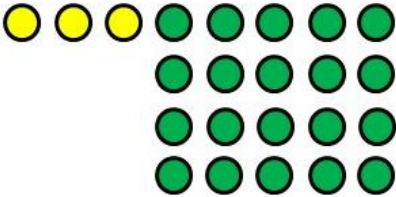
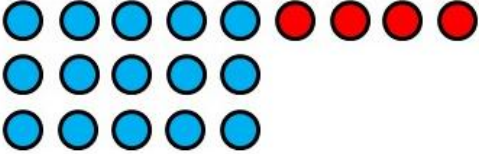
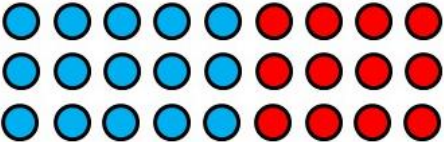
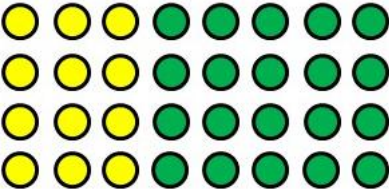
11.1.2 Collaboration: Modeling the Meaning of Expressions

1. Work with your partner to determine which mathematical model matches each numerical expression. Use arrows to indicate matches. Be prepared to explain your thinking.

Numerical Expression	Mathematical Model
$(3 \times 5) + 4$	
$3 \times (5 + 4)$	
$(3 + 5) \times 4$	
$3 + (5 \times 4)$	

2. How did you use the grouping symbols to help you determine which model matched each expression?

3. If the grouping symbols were removed, which model matches each expression? Use arrows to indicate matches.

Numerical Expression	Mathematical Model
$3 \times 5 + 4$	
$3 + 5 \times 4$	
	
	

11.1.3 Guided Instruction: Applying Order of Operations to Algebraic Expressions

Recall from 5th grade that an order of operations is used to evaluate numeric expressions with multiple operations. Grouping symbols such as parentheses are used to indicate which operations should be performed first. The following order is used.

- Step 1: Perform operations within grouping symbols.
- Step 2: Multiply and divide in order from left to right (→).
- Step 3: Add and subtract in order from left to right (→).

1. Three students completed the warm-up. The steps in each student’s process are shown.

Leila’s Work	Fabian’s Work	Jaylah’s Work
$4 - 2 \times 6 \div 3 + 7$ $2 \times 6 \div 3 + 7$ $12 \div 3 + 7$ $4 + 7$ 11	$4 - 2 \times 6 \div 3 + 7$ $4 - 12 \div 3 + 7$ $4 - 4 + 7$ $4 - 11$ -7	$4 - 2 \times 6 \div 3 + 7$ $4 - 12 \div 3 + 7$ $4 - 4 + 7$ $0 + 7$ 7

- a. Complete the statements about which student correctly applied the given steps of the order of operations.
- ☐ Leila

☐ Fabian

☐ Jaylah

evaluated the numerical expression correctly.

She first performed all

☐ multiplication and division

☐ addition and subtraction

in

order from left to right and then performed all

☐ multiplication and division

☐ addition and subtraction

in order from left to right.

- b. With your partner, discuss the work of each student, Leila, Fabian, and Jaylah. Two of these students made a mistake somewhere. Circle the mistakes that you and your partner found.

2. Omari evaluated Expression A and Expression B. His work is shown.

Expression A	Expression B
$12 + 24 \div 6 - 2 \times 2$ $12 + 4 - 2 \times 2$ $12 + 4 - 4$ $16 - 4$ 12	$12 + 24 \div (6 - 2 \times 2)$ $12 + 24 \div (6 - 4)$ $12 + 24 \div (2)$ $12 + 12$ 24

- a. With your partner, discuss the two original expressions that Omari evaluated and his corresponding answers.
- b. Omari's partner gave him a 👍 on both of his answers. Why did Omari arrive at two different values when evaluating these expressions?

Like numeric expressions, algebraic expressions can be evaluated using order of operations when given a specific value for the **variable(s)**.

In an algebraic expression, a **variable** is a letter that represents a number. It is called a **variable** because the value used to replace the letter can **vary**.

3. When evaluating algebraic expressions, first substitute the given _____ for the variable(s). Then, apply the given steps for order of operations.



In mathematics, **evaluate** is an instruction that indicates the answer is a **value**.

4. Evaluate $\frac{n}{6} + 2$ when $n = 12$.

In numeric expressions, multiplication can be represented in different ways. Each expression below is a way to represent “three times five.”

$$3 \times 5$$

$$3 \cdot 5$$

$$3(5)$$

In algebraic expressions, the multiplication symbol, $\times$, could be confused with the variable x . Therefore, writing a number next to a variable is a better way to note multiplication when using variables.

$3x$ means “three times x .”

Multiplication of a number and a variable could also be represented using the multiplication dot, $3 \cdot x$, or parentheses, $3(x)$, but $3x$ will be the most common way.

5. Explain why $3 \cdot 5$ cannot be written as 35.
6. What is the value of $3x$ if $x = 5$?



When substituting values into algebraic expressions, it is helpful to put the substituted value in **parentheses** when the variable is being multiplied by a known value.

7. Find the value of $7m - (3 + b - 12) \div 2$ if $m = 4$ and $b = 15$. Show your work.

11.1.4 Guided Practice: Evaluating Algebraic Expressions

1. What is the value of the expression $6 \cdot 2 - (t + 3)$ if $t = 5$?

2. Find the value of $4 - 10(5 - w) + 4w \cdot 2$ when $w = 3$.

3. Evaluate $6p + \frac{8}{H} - 7$ for $p = 4$ and $H = 2$.

1. Evaluate $13 - 48 \div 2r + 9$ if $r = 8$.

2. Find the value of $6 \cdot (q - 3q + 9)$ when $q = 7$.

3. What is the value of $7 + \frac{4}{a} - \frac{b}{3} \times 5$ if $a = 1$ and $b = 9$?

11.1.6 Wrap-Up: Ticket Out the Door

1. Evaluate $8a + 3b(2a - 4)$ for $a = 2$ and $b = 6$.